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Ledger-based provenance for peer review: a verifiable record of editorial decisions

Evidence from 12,400 manuscripts across open-access venues

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ABSTRACT

Peer review remains the principal quality signal in scholarly communication, yet the record of how a decision was reached is rarely available to readers. We introduce a ledger-based provenance model in which every editorial transition — submission, reviewer assignment, review round, correction, and final decision — is recorded as an append-only event bound to the published article. Applying the model to 12,400 manuscripts, we show that verifiable provenance reduces post-publication disputes by 38% and shortens median time-to-decision by 11 days, without increasing editorial workload. We further show that exposing a decision trail is compatible with double-anonymised review, since provenance can be published at the level of events rather than identities. The approach is implemented as an open specification and is independent of any single publisher or platform.

Keywords: peer review; provenance; open science; editorial workflow; research integrity; scholarly infrastructure

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1. Introduction

The credibility of a published result rests less on the result itself than on the process that produced it. Readers are asked to trust that a manuscript was examined by competent referees, that criticism was addressed, and that the final decision followed from the evidence rather than from convenience. In practice this process is opaque: the reader sees an outcome and must infer the reasoning behind it.

Several initiatives have attempted to open parts of this record — published referee reports, transparent decision letters, registered reports. Each addresses a symptom, but none produces a machine-verifiable account of the editorial process as a whole. What is missing is not more disclosure, but a *structure* for it.

A decision trail that cannot be verified is a claim, not a record.

2. Materials and Methods

2.1 Data sources

We analysed 12,400 manuscripts submitted between 2021 and 2025 across seven open-access venues. Each manuscript contributed a complete editorial event stream, from initial submission to final decision, comprising 184,930 discrete transitions in total.

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Figure 1. Editorial event stream for a representative manuscript. Each node is an append-only transition; shaded regions mark review rounds. Identities are withheld while the structure of the decision remains fully auditable.

2.2 Provenance model

Every transition is recorded as an immutable event carrying an actor role (not identity), a timestamp, the prior and resulting states, and a content digest of any attached artefact. Events are chained, so any retroactive edit invalidates all subsequent links and is detectable by a third party.

Table 1. Outcome measures before and after adoption of ledger-based provenance.

Measure	Before	After	Δ
Median time to first decision (days)	63	52	–11
Post-publication disputes (per 1,000)	8.4	5.2	–38%
Reviewer response rate	41%	47%	+6 pp
Editorial hours per manuscript	4.1	4.0	–0.1

3. Results

Adoption was associated with a reduction in median time-to-decision from 63 to 52 days. The effect was concentrated in the reviewer-assignment phase, where visibility of prior invitation attempts reduced duplicated outreach. Editorial workload was unchanged within measurement error, indicating that the gains came from coordination rather than from additional effort.

4. Discussion

The central finding is that transparency and anonymity are not in tension when provenance is published at the level of events. Readers gain a verifiable account of process; referees retain the protection that candid criticism requires.

5. Conclusions

A verifiable editorial record is achievable with existing infrastructure and modest process change. We release the specification openly so that it can be adopted independently of any single platform.

AUTHOR CONTRIBUTIONS

Conceptualisation, L.F.C. and M.A.R.; methodology, L.F.C.; formal analysis, T.N.; data curation, H.V.; writing — original draft, L.F.C.; writing — review & editing, all authors. All authors have read and agreed to the published version of the manuscript.

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DATA AVAILABILITY

The event-stream dataset and analysis code are openly available at <https://doi.org/10.5281/zenodo.0000000>.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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VERIFIED EDITORIAL RECORD

Record digest · a7f3c9e1b482...d05f



Verify at **sciledger.org/verify/2026.118**
Review rounds **2 · 4 referee reports**
Decision **Accepted 19 Mar 2026 by the handling editor**
Events recorded **31 transitions · append-only**

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